

Frequency Adapter with SAM for Generalized Medical Image Segmentation

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Abstract. Medical image segmentation is a critical task in computer-aided diagnosis and treatment planning. However, deep learning models often struggle to generalize across datasets due to domain shifts arising from variations in imaging protocols, scanner types, and patient populations. Traditional domain generalization (DG) methods utilize causal feature learning, adversarial consistency, and style augmentation to improve segmentation robustness. While effective, these approaches rely on explicit feature alignment, adversarial objectives, or handcrafted augmentations, which may not fully exploit the capabilities of foundation models. Recently, the Segment Anything Model (SAM) has demonstrated strong generalization capabilities in segmentation tasks. SAM-based DG methods attempt to improve medical image segmentation. However, these approaches primarily operate in the spatial domain and overlook frequency-based discrepancies that significantly affect model robustness. In this work, we propose Frequency-guided Domain Generalization with SAM (FreeSAM), a novel framework that integrates Low-Rank Adaptation (LoRA) for efficient fine-tuning and a frequency adapter to incorporate frequency-domain representations for single-source domain generalization. FreeSAM enhances SAM’s segmentation robustness by extracting domain-invariant high-frequency features, mitigating frequency-related domain shifts. Experimental results on fundus and prostate datasets demonstrate that FreeSAM outperforms existing traditional DG and SAM-based DG approaches in domain generalization.

Keywords: Domain generalization · Frequency domain · SAM.

1 Introduction

Medical image segmentation plays a crucial role in computer-aided diagnosis and treatment planning [1, 8, 22]. However, deep learning models often fail to generalize across datasets from different institutions, imaging devices, and patient populations due to domain shifts [12]. Domain generalization (DG) aims to address this issue by training models to perform well on unseen distributions

without requiring access to target domain data during training. Unlike domain adaptation, which assumes access to unlabeled target data for fine-tuning, DG forces models to learn robust and transferable representations that remain effective in diverse environments. Existing DG approaches can be categorized into traditional DG methods and SAM-based DG methods.

Traditional DG methods enhance segmentation robustness through causal feature learning, adversarial consistency, and style augmentation. CSDG [19] extracts domain-invariant features to mitigate distribution shifts, while ACL [23] and contrastive learning [11] enforce feature alignment and consistency across different domains. Style-based approaches such as MixStyle [26] and MaxStyle [3] augment training by blending styles from multiple domains, encouraging models to learn more generalizable representations. However, these approaches rely on explicit feature alignment, adversarial objectives, or handcrafted augmentations, which may not fully leverage foundation models like SAM.

The Segment Anything Model (SAM) [13], trained on over 1 billion masks, has revolutionized segmentation tasks with its remarkable generalization capabilities. Researchers have adapted SAM for medical image segmentation, with methods like Decoupled SAM (DeSAM) [8] improving robustness by separating domain-specific and domain-agnostic features, and DAPSAM [22] utilizing prototype-based prompts to enhance segmentation in unseen domains. However, these SAM-based approaches predominantly operate in the spatial domain, overlooking frequency-based domain shifts that significantly influence medical imaging due to variations in texture, contrast, and resolution. Recent theoretical insights [14, 15] reveal that deep neural networks exhibit biases toward specific frequency components during learning, which can undermine feature robustness and exacerbate domain shift challenges.

In this work, we propose Frequency-guided Domain Generalization with SAM (FreeSAM), a pioneering framework that synergistically integrates Low-Rank Adaptation (LoRA) for efficient fine-tuning of SAM with a novel Frequency Adapter to incorporate frequency-domain features, significantly enhancing single-source domain generalization. FreeSAM leverages SAM’s feature extraction capabilities while employing the Frequency Adapter (FA) to capture high-frequency, domain-invariant structures, effectively addressing frequency-related domain shifts. These features are seamlessly decoded into precise segmentation masks using an automated prompt generator [22]. Our contributions are detailed as follows:

- We propose FreeSAM, a novel framework that leverages frequency-guided adaptation of the Segment Anything Model to achieve superior domain generalization in medical image segmentation.
- We develop an innovative Frequency Adapter that effectively captures and integrates high-frequency components, significantly enhancing robustness against frequency-related domain shifts.
- Extensive experiments on fundus and prostate datasets demonstrate that FreeSAM markedly outperforms both traditional domain generalization techniques and existing SAM-based approaches, setting a new benchmark for segmentation robustness.

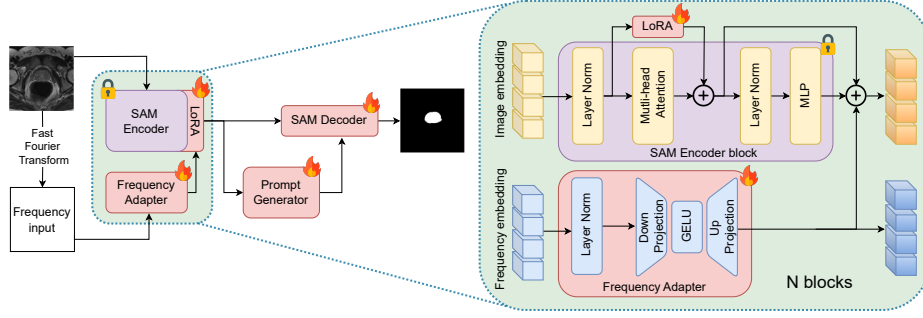


Fig. 1. Overview of the proposed frequency-guided domain generalization framework with SAM (FreeSAM). The fire icon represents trainable parameters, while the lock icon indicates frozen parameters retained from the pre-trained model.

2 Method

In this section, we first define the single-source domain generalization (SSDG) problem. Then, we elaborate on the proposed FreeSAM method. Finally, we detail the overall training objective. The SSDG is defined as training on a single source domain and then testing model performance on unseen test domains. SSDG considers a scenario where a model is trained using data from a single labeled source domain $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$, where x_i^s represents medical images and y_i^s denotes the corresponding segmentation masks. The objective is to learn a segmentation model f_θ parameterized by θ such that it performs well on an unseen target domain $\mathcal{D}_t$, where $\mathcal{D}_s \neq \mathcal{D}_t$ and no target domain samples are available during training. As shown in Figure 1, we adapt SAM [13] using LoRA [9] and a frequency adapter for single-source domain generalization.

2.1 FreeSAM encoder

Given the input image I of dimension $H \times W$, we obtain the frequency input using Fourier transform. These inputs are processed using SAM’s encoder and FA module to leverage the strengths of both spatial and frequency domains.

Frequency Adapter The Discrete Fourier Transform (DFT) is a widely used technique to convert an image from the spatial to the frequency domain. The Fast Fourier Transform (FFT) is practically employed for efficient computation of DFT, the spectrum representation of $f(h, w)$ can be formulated as:

$$F(u, v) = \sum_{h=0}^{H-1} \sum_{w=0}^{W-1} f(h, w) \cdot e^{-j2\pi(\frac{uh}{H} + \frac{vw}{W})} \quad (1)$$

where $f(h, w)$ represents the pixel intensity at spatial coordinates (h, w) in the image domain, $F(u, v)$ is the corresponding frequency spectrum at frequency coordinates (u, v) . The exponential term captures the oscillatory basis functions

used to decompose the image into its frequency components. The amplitude and phase components of $F(u, v)$ are derived as $|F(u, v)|$ and $\arg(F(u, v))$, respectively. To efficiently compute the Discrete Fourier Transform, we utilize FFT in PyTorch to reduce the computational complexity from $\mathcal{O}(HW^2)$ in the naive DFT implementation to $\mathcal{O}(HW \log HW)$, making it feasible for medical image.

Next, both the original input and its amplitude (or phase) component in the frequency domain are processed through the Patch Embedding layer of the ViT architecture [7], generating the image and frequency embeddings, respectively. As illustrated in Fig. 1, the FA consists of a linear down-projection layer, followed by a GELU activation function, and a linear up-projection layer. To fully integrate frequency-domain information, we employ N frequency adapters, where N corresponds to the number of ViT blocks in SAM’s encoder.

Low-Rank Adaptation (LoRA) [9] introduces trainable low-rank matrices into the attention layers, enabling efficient adaptation while maintaining the pre-trained knowledge of the foundation model. Specifically, instead of updating the full weight matrix $W \in \mathbb{R}^{d \times d}$ in the self-attention layers of SAM, LoRA parameterizes the weight update as:

$$W' = W + \Delta W, \quad \text{where} \quad \Delta W = AB$$

where $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times d}$ are learnable low-rank matrices, and $r \ll d$ is the rank of the adaptation. This formulation significantly reduces the number of trainable parameters while preserving the expressiveness of the model. As observed in [25], the multi-head self-attention mechanism leverages cosine similarity to identify regions of focus, making it effective to apply LoRA to the query and value projection layers to modulate attention scores and enhance model efficiency. Therefore, LoRA-updated query and value projections are:

$$\begin{aligned} W'_Q &= W_Q + A_Q B_Q \\ W'_V &= W_V + A_V B_V \end{aligned}$$

where W_Q and W_V are the frozen projection layers inherited from SAM’s image encoder, while A_Q , B_Q , A_V , and B_V represent the trainable LoRA parameters. By freezing the pre-trained SAM parameters and updating only the LoRA layers, we enable efficient fine-tuning for medical image segmentation while significantly reducing computational overhead. At each stage of SAM’s encoder, the output from the ViT block is combined with the corresponding output from the frequency adapter before being propagated to the next ViT block. The final encoder output is then utilized by SAM’s decoder and the prototype-based prompt generator [22], which is further detailed in the following section.

2.2 Prompt generator

This module, proposed by Wei et al. [22], employs a memory-based prototype learning approach to enhance domain adaptability. Initially, global average pooling and max pooling are applied to the FreeSAM encoder output e_i to derive an

instance-level prototype p_i . The memory bank is structured as a parameterized matrix $M \in \mathbb{R}^{N \times C}$, where N represents the number of stored prototypes, and C denotes their dimensionality. Given a prototype vector $p_i \in \mathbb{R}^{1 \times C}$ extracted from an image embedding, the memory bank retrieves and updates relevant prototypes to generate a domain-adaptive representation $\tilde{p}_i$. The similarity between p_i and stored prototypes in M is computed using cosine similarity:

$$s_i = \frac{p_i M^T}{\|p_i\| \|M\|}$$

The refined prototype $\hat{p}_i$ is computed via weighted aggregation as follows:

$$\hat{p}_i = \sum_{j=1}^N \alpha_j M_j, \quad \text{where} \quad \alpha_j = \frac{\exp(s_{i,j})}{\sum_{k=1}^N \exp(s_{i,k})}$$

where $\hat{p}_i$ and α_j represent the refined prototype and the attention weight assigned to each prototype M_j . The memory bank dynamically updates throughout training to preserve prototype consistency across different domains. To further refine the embedding, Wei et al. [22] compute the cosine similarity between $\hat{p}_i$ and FreeSAM’s encoder output, generating an activation map A_i as guidance information. Finally, the refined prototype $\hat{p}_i$, encoder embedding e_i , and activation map A_i are concatenated and processed through a 1×1 convolutional layer to generate the final prompt for SAM’s decoder.

Training objective: Following SAMed [25] and TriD [5], we employ a hybrid loss function that combines CE loss and Dice loss to optimize the segmentation model in the source domain. The overall objective function is as follows:

$$\mathcal{L} = (1 - \lambda) \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{Dice}}, \quad (2)$$

where $\mathcal{L}_{\text{CE}}$ is the standard pixel-wise cross-entropy loss, $\mathcal{L}_{\text{Dice}}$ is the Dice loss for region-based supervision, and λ is a weighting factor that balances the contribution of both loss terms, which is set to 0.8.

3 Experiments

3.1 Experimental setup and implementation details

Dataset: To evaluate the performance of the proposed FreeSAM, we conducted experiments in two settings: The first dataset is RIGA+ dataset [6, 10], which is a multi-domain benchmark for joint optic cup (OC) and optic disc (OD) segmentation, comprising annotated fundus images from five distinct domains: BinRushed, Magrabia, Messidor Base 1, Messidor Base 2, and Messidor Base 3. For our segmentation experiments, we use BinRushed and Magrabia as the source domains for training, while the remaining three domains are treated as target domains to evaluate cross-domain generalization. We adopt the official dataset split. To ensure consistency across different domains, all images are resized to 512×512 . Model performance is evaluated using the Dice score (DSC).

Table 1. Segmentation performance on the multi-site RIGA+ fundus dataset. The best and second-best results are highlighted in **bold** and underlined, respectively. The table is divided into two parts: the upper part corresponds to experiments using BinRushed as the source domain, while the lower part represents results obtained with Magrabia as the source domain.

Method	Venue/Year	Encoder	Messidor	Base 1	Messidor	Base 2	Messidor	Base 3	Average	
			OD	OC	OD	OC	OD	OC	OD	OC
BinRushed										
CSDG [19]	TMI'2022	UNet	93.56	81.00	94.38	83.79	93.87	83.75	93.93	82.85
ADS [23]	MICCAI'2022		94.07	79.60	94.29	81.17	93.64	81.08	94.00	80.62
MaxStyle [3]	MICCAI'2022		94.28	82.61	86.65	74.71	92.36	82.33	91.09	79.88
D-Norm [27]	CVPR'2022		94.57	81.81	93.67	79.16	94.82	83.67	94.35	81.55
SLAug [21]	AAAI'2023		95.28	83.31	95.49	81.36	95.57	84.38	95.45	83.02
CCSDG [11]	MICCAI'2023		95.73	86.13	95.73	86.28	95.45	86.77	95.64	86.57
SAMed [25]	arXiv'2023	ViT	95.28	84.24	94.11	80.21	94.84	82.60	94.74	82.35
DeSAM [8] [b]	MICCAI'2024		89.33	79.68	93.44	82.97	91.51	82.70	91.42	81.78
DeSAM [8] [p]	MICCAI'2024		91.79	80.87	92.57	82.95	93.66	84.19	92.67	82.67
DAPSAM [22]	MICCAI'2024		96.34	88.24	<u>96.10</u>	86.31	96.34	88.77	96.26	87.77
FreeSAM	—		<u>96.13</u>	<u>88.12</u>	96.29	<u>86.20</u>	<u>96.32</u>	<u>88.28</u>	<u>96.25</u>	<u>87.53</u>
Magrabia										
CSDG [19]	TMI'2022	UNet	89.67	75.39	87.97	76.44	89.91	81.35	89.18	77.73
ADS [23]	MICCAI'2022		90.75	77.78	90.37	79.60	90.34	79.99	90.48	79.12
MaxStyle [3]	MICCAI'2022		91.63	78.74	90.61	80.12	91.22	81.90	91.15	80.25
D-Norm [27]	CVPR'2022		92.35	79.02	91.23	80.06	92.09	79.87	91.89	79.65
SLAug [21]	AAAI'2023		93.08	80.70	92.70	80.15	92.23	80.89	92.67	80.58
CCSDG [11]	MICCAI'2023		94.78	84.94	95.16	85.68	95.00	85.98	94.98	85.53
SAMed [25]	arXiv'2023	ViT	95.41	85.26	95.36	84.25	95.38	84.76	95.38	84.76
DeSAM [8] [b]	MICCAI'2024		82.45	69.66	84.97	75.75	83.86	74.74	83.76	73.38
DeSAM [8] [p]	MICCAI'2024		81.39	67.88	83.95	76.33	79.99	73.05	84.50	72.42
DAPSAM [22]	MICCAI'2024		<u>96.22</u>	<u>86.74</u>	<u>96.32</u>	<u>89.59</u>	<u>96.35</u>	<u>88.12</u>	<u>96.30</u>	<u>88.15</u>
FreeSAM	—		96.28	87.61	96.41	90.07	96.48	88.20	96.39	88.63

The second dataset is the prostate dataset [16], which consists of 116 cases spanning six different domains: A (RUNMC), B (BMC), C (I2CVB), D (UCL), E (BIDMC), and F (HK). All slices are resized to 384×384 with standardized voxel spacing for consistency. We follow the preprocessing pipeline of MaxStyle [3]. For each experiment, one domain is designated as the source domain, where the data is split into training (90%) and in-domain validation (10%). The trained model is then evaluated on all other unseen domains to assess cross-domain generalization.

Implementation details: We configure the Low-rank Adaptation (LoRA) rank to 4, balancing efficiency and performance. The frequency adapters incorporate amplitude or phase information to enhance feature extraction. Training is conducted using the ViT-B variant of SAM, with an initial learning rate of $5e-4$ and weight decay of 0.1 for the AdamW optimizer [17]. Following SAMed [25], we adopt a warm-up strategy, setting warm-up periods to 250 for the prostate dataset and 25 for RIGA+, considering their differing data configurations. Training is early stopped at 160 epochs, with a maximum of 200 epochs, and the loss weighting factor is set to 0.8. The code and pre-trained models will be released on GitHub for reproducibility.

Table 2. Segmentation performance on multi-site prostate dataset. The best and second-best results are highlighted in **bold** and underline, respectively. Each column represents leave-one-out results for the model trained on the corresponding domain while testing on the other domains.

Method	Year	Encoder	A	B	C	D	E	F	Average
In-domain	–	UNet	85.38	83.68	82.15	85.21	87.04	84.29	84.63
AdvBias [4]	MICCAI’2020	UNet	77.45	62.12	51.09	70.20	51.12	50.69	60.45
RandConv [24]	arXiv’2020		75.52	57.23	44.21	61.27	49.98	54.21	57.07
MixStyle [26]	arXiv’2021		73.04	59.29	43.00	62.17	53.12	50.03	56.78
MaxStyle [3]	MICCAI’2022		81.25	70.27	62.09	58.18	70.04	67.77	68.27
CSDG [19]	TMI’2022		80.72	68.00	59.78	72.40	68.67	70.78	70.06
SAMed [25]	arXiv’2023	ViT	80.42	<u>81.44</u>	66.75	82.09	80.19	80.17	78.51
DeSAM [8] [b]	MICCAI’2024		82.30	78.06	66.65	82.87	77.58	79.05	77.75
DeSAM [8] [p]	MICCAI’2024		82.80	80.61	64.77	83.41	80.36	<u>82.17</u>	79.02
MedSAM [18]	MICCAI’2024		72.32	73.31	61.53	64.46	68.89	61.39	66.98
DAPSAM [22]	MICCAI’2024		86.34	81.05	<u>70.81</u>	<u>85.28</u>	<u>82.91</u>	81.48	<u>81.31</u>
FreeSAM	–		<u>86.16</u>	82.52	74.98	86.44	83.14	83.19	82.74

3.2 Performance evaluation

RIGA+ dataset: As presented in Table 1, FreeSAM demonstrates superior performance compared to traditional and SAM-based DG methods. When using Magrabia as the source domain, FreeSAM outperforms the CNN-based state-of-the-art CCSDG, achieving a 1.41% improvement in optic disc segmentation (96.39% vs. 94.98%) and a 3.10% gain in optic cup segmentation (88.63% vs. 85.53%). Additionally, FreeSAM attains competitive or superior performance compared to other SAM-based methods, further validating its effectiveness.

Prostate dataset: Compared to the traditional CNN-based U-Net architecture [20], ViT-based methods built on SAM demonstrate superior segmentation performance, as shown in Table 3. Our proposed method not only surpasses the best-performing CNN-based approach but also outperforms recent SAM-based methods on the prostate dataset. Specifically, compared to CSDG, the top-performing CNN-based method, FreeSAM achieves a significant 12.68% improvement, highlighting the effectiveness of incorporating frequency-aware adaptation. Furthermore, our method outperforms the recently proposed SAM-based approach, DAPSAM [22], achieving an additional 1.43% performance gain, demonstrating the advantage of integrating LoRA fine-tuning and frequency-domain representations for robust medical image segmentation.

3.3 Ablation study

Computational complexity We conducted experiments on a single A6000 GPU using the RUNMC prostate dataset (384×384 images). FreeSAM, integrating LoRA and a Frequency Adapter (FA), substantially lowers computational demands compared to full fine-tuning [2]. In particular, FreeSAM achieves lower

FLOPS (113.5G vs. 117.2G), fewer learnable parameters (8.47M vs. 15.37M), and fewer total parameters (96.97M vs. 101.86M) than DAPSAM, while improving Dice scores to 82.74% from 80.12%. These results highlight FreeSAM’s superior efficiency and accuracy, ideal for real-world medical imaging.

Frequency component analysis We evaluated our FreeSAM framework’s FA on fundus and prostate datasets. As shown in Table 3, employing FA to inject frequency information consistently improves segmentation performance, e.g. at least an improvement of 2.11% Dice score on the prostate dataset. In terms of frequency components, amplitude-based FA (FA_{amp}) achieves higher optic disc/optic cup Dice scores (96.25%/87.53%) on the RIGA+ dataset with Bin-Rushed as source domain (RIGA_{Bin}) than phase-based FA (FA_{pha} , 96.13%/86.73%).

Table 3. Ablation study on frequency components. The best Dice scores are highlighted in **bold**, respectively.

Settings	Backbone	Prostate	RIGA_{Bin}		RIGA_{Mag}	
			Optic disc	Optic cup	Optic disc	Optic cup
w/o FA	ViT	79.58	95.78	84.96	95.02	86.12
FA_{pha}		81.69	96.13	86.73	95.95	87.91
FA_{amp}		82.74	96.25	87.53	96.39	88.63

4 Conclusion

In this work, we propose FreeSAM, a novel framework designed to enhance the generalization capability of SAM for medical image segmentation. FreeSAM leverages LoRA for efficient fine-tuning and integrates a FA module to extract domain-invariant frequency features, effectively mitigating domain shifts by incorporating frequency-domain representations. Comprehensive experiments on fundus and prostate datasets demonstrate that FreeSAM consistently outperforms both traditional domain generalization methods and SAM-based DG approaches, achieving superior segmentation accuracy on unseen domains. These results highlight the efficacy of combining frequency-aware adaptation with LoRA-based fine-tuning in foundation models to improve cross-domain robustness.

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